Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

Sea Ice Thematic Assembly Centre

SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_002

SEAICE_ANT_PHY_AUTO_L3_NRT_011_012

Issue: 2.0

Contributors: Frode Dinessen and Cecilie Wettre (MET), Matilde B. Kreiner and Tore Wulf (DMI)

Approval date by the CMEMS product quality coordination team: 14/11/2025

CHANGE RECORD

When the quality of the products changes, the QUID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies. The validation (see last column in the table) means the document was checked for contents, formats and consistency with other documents.
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Issue	Date	§	Description of Change	Author	Validated By
1.0	01.09.2023	All	Creation of the document, Based on a split of v2.17	Anton Korosov, Cecilie Wettre, Matilde Kreiner, Frode Dinessen	C. Wettre
1.1	01.09.2024	All	Replace dataset METNO-ARC-SEAICE_CONC-L4-NRT-OBS. The DMI-ASIP sea ice classification dataset (cmems_obs-si_arc_phy_nrt_l3_P1D) has been removed, since the dataset will be retired. Sea Ice type (cmems_obs-si_arc_phy-icetype_nrt_L4-auto_P1D): Both NIC and DMI ice charts are used for validation and the validation process used data from 1 January 2021 to April 2024.	Frode Dinessen, Matilde Kreiner, Anton Korosov, Mohamed Babiker	C.Wettre
2.0	14.07.2025		Removed textual references to retired product SEAICE_ARC_PHY_AUTO_L4_NRT_011_015	C. Wettre	C. Wettre

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ACRONYM TABLE

AI	Artificial Intelligence
AMSR2	Advanced Scanning Microwave Radiometer 2
ASIC	Automatic Sea Ice Concentration
ASIP	Automated Sea Ice Products
AVHRR	Advanced Very High Resolution Radiometer
CMEMS	Copernicus Marine Environment Monitoring Service
CF	Climate Forecast (convention for NetCDF)
DL	Deep Learning
DMI	Danish Meteorological Institute
(I)CDR	(Interim) Climate Data Record
EUMETSAT	European Organisation for the Exploitation of Meteorological Satellites
EW	Extra-Wide Swath
FLZ	Floe Size
FORECAST	An assessment of the future state of the ocean variables made using numerical models.
FTP	Protocol to download files
GLO	Global
GRDM	Ground Range Detected Multilooked
HH	Horizontal transmit, Horizontal receive SAR polarization
HV	Horizontal transmit, Vertical receive SAR polarization
IC	Ice Concentration
L3	Level-3
L4	Level-4
MSIC	Manually derived ice chart
MET	Norwegian Meteorological Institute
MDS	Marine Data Store
MFC	Monitoring and Forecasting Centre
MY	Multi Year
NERSC	Nansen Environmental Remote Sensing Centre

NetCDF	Network Common Data Form
NOAA	National Oceanic and Atmospheric Administration
NRT	Near Real Time
OSI SAF	Ocean & Sea Ice Satellite Application Facility
PUM	Product User Manual
QUID	Quality Information Document
SAR	Synthetic Aperture Radar
SI	Sea Ice
SIC	Sea Ice Concentration
SSMIS	Special Sensor Microwave Imager / Sounder
SOD	Stage-of-development
SST	Sea Surface Temperature
TAC	Thematic Assembly Center
VIIRS	Visible Infrared Imaging Radiometer Suite
WMO	World Meteorological Organization

KEY DEFINITIONS

Sea Surface Temperature	The temperature of the ocean water close to the surface.
Sea Ice Concentration	Fraction (percentage) of the sea surface area of the grid cell containing sea-ice.
Level-2 (L2) data product	Geophysical variables derived from Level 1 source data on the Level 1 grid (typically the satellite swath projection).
Level-3 (L3) data product	L2 data granules remapped to a regular latitude/longitude grid. In this project, a typical L3 file may contain all the observations from one or more instruments in a 24-hour period.
Level 4 (L4) data product	Observations from multiple instruments using an analysis system (e.g., optimal interpolation) to produce a gridded, gap-free product.

RELEVANT DOCUMENT LIST

CMEMS-SEAICE-SQO-011-002-012	The Synthesis Quality Overview document is a short summary of the quality information presented here. It can be found in the dedicated product page[1] under Documentation
CMEMS-SEAICE-PUM-011-002	The Product User Manual presents the product details along with file structure. It can be found in the dedicated product page under Documentation.
CMEMS-SEAICE-PUM-011-012	The Product User Manual presents the product details along with file structure. It can be found in the dedicated product page[2] under Documentation.

[1] See https://data.marine.copernicus.eu/product/SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_002/description

[2] See https://data.marine.copernicus.eu/product/SEAICE_ANT_PHY_AUTO_L3_NRT_011_012/description

I EXECUTIVE SUMMARY

I.1 Products covered by this document

This document provides information to Copernicus Marine Service (CMEMS) users regarding the quality of manually (for Svalbard and Greenland regions) and automatically generated high resolution sea ice concentration (for the Antarctic). The products are delivered daily through the Dissemination Unit and the product names are listed in Table 1 below. Coverage of the products is displayed in Figures 1 and 2.

Table 1: SI TAC Sea Ice products and partner roles.

Product	***Product description***	***Production unit (PU)***	***Production Centre (PC)***
SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_002	Arctic Ocean - sea Ice L4 sea ice concentration Charts - Svalbard and Greenland	SIW-METNO-OSLO-NO; SIW-DMI-COPENHAGEN-DK	SIW-METNO-OSLO-NO
SEAICE_ANT_PHY_AUTO_L3_NRT_011_012	Antarctic high-resolution ice product	SIW-DMI-COPENHAGEN-DK	SIW-METNO-OSLO-NO

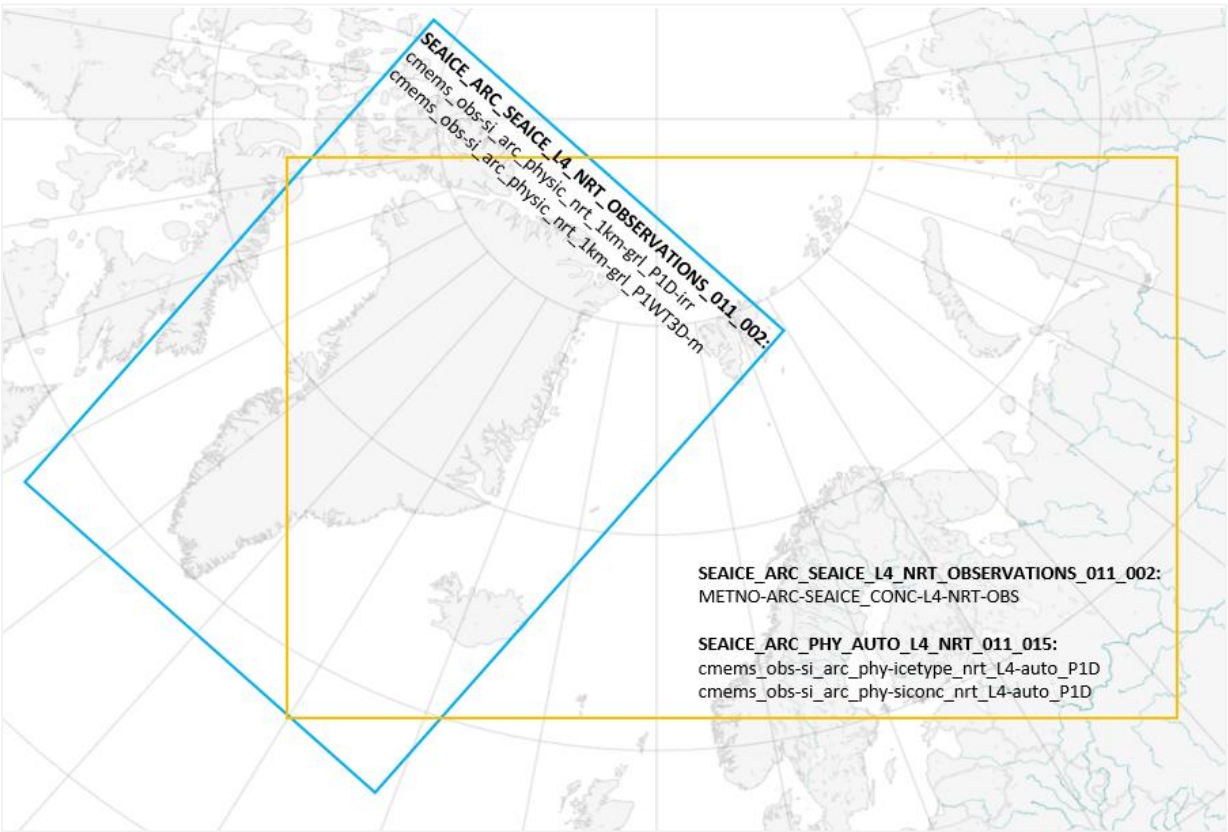


Figure 1: Coverage of the Arctic sea ice chart datasets for the product 011_002: The Greenland ice charts (blue bounding box) and the Svalbard ice chart (yellow bounding box).

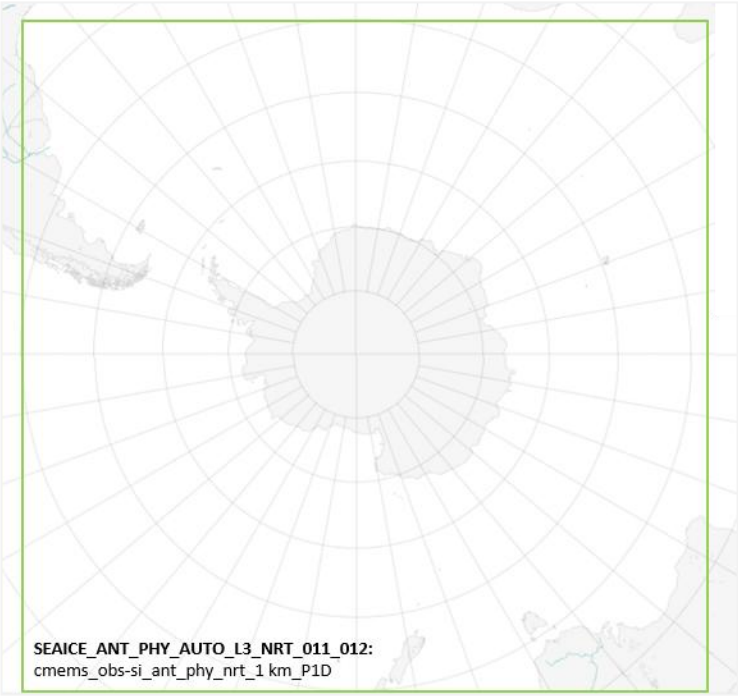


Figure 2: Coverage of the Antarctic dataset for the products 011_012.

I.2 Summary of the results

In contrast to other satellite measurements, such as Sea Surface Temperature (SST) and wind, for the present products there is a significant lack of useful in-situ observations with which to compare the satellite data. This is true for all the variables provided in the sea ice products: concentration, ice edge, ice type, ice drift, iceberg concentration, and ice surface temperature. The in-situ observations that are available are sparse and intermittent in time. It is therefore very difficult to calculate reliable statistics for the global datasets and nearly impossible for the regional datasets. For this reason, a common validation method for the different products has not been established and validation is not following conventions laid out in the Copernicus Marine Environment Monitoring Service (CMEMS) Calibration/Validation (Cal/Val) guidelines. In the following section we provide a short summary of the validation performed for the individual products. A more detailed description can be found in the validation section (section III).

I.2.1 Arctic Ocean – Sea Ice Concentration Charts – Svalbard and Greenland

SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_002

The sea ice concentration ice charts from the national Ice Services at the Norwegian Meteorological Institute (MET) and the Danish Meteorological Institute (DMI) are generated based on a manual interpretation of high-resolution satellite data. The main sensor used is data from the Synthetic

Aperture Radar (SAR) sensor on-board Sentinel-1 and Radarsat-2, due to the SAR sensors´ high nominal spatial resolution and lack of dependence on the weather.

The ice concentration charts from the national Norwegian Ice Service are produced every weekday and have a fixed geographical coverage from East Greenland to Novaya Zemlya, east of the Barents Sea. The ice concentration charts from the Danish national ice service are produced in 2 variants: the Regional Near Real Time (NRT) ice chart, which typically covers an approximately 500 x 500 km subset of the area observed by a radar satellite, and the Overview ice chart, which is produced twice per week and covers the entire area around Greenland. The nominal quality of the charts is similar, and validation procedures are the same. Fore reference, a study and discussion of the ice chart uncertainty is included in the Appendix.

The limited number of in-situ measurements available in the Arctic makes it very difficult to give a quantitative number of the accuracy for the ice chart products. In addition, it is difficult to quantify the subjectivness in the manual concentration estimates. Use of optical high resolution data from Sentinel-2 and the Visible Infrared Imaging Radiometer Suite (VIIRS) sensors onboard National Oceanic and Atmospheric Administration (NOAA) satellites are used for a visual comparison, but this is also difficult to quantify. In the validation performed on data from 2013 we used some optical satellite data from the NOAA satellites and an alternative SAR source from the COSMO SkyMed satellite[3]. The data were visually compared against the ice charts.

The products are compared against the European Organisation for the Exploitation of Meteorological Satellites (EUMETSAT) Ocean and Sea Ice Satellite Application Facility (OSI SAF) ice concentration product (SEAICE_GLO_SEAICE_L4_NRT_OBSERVATIONS_011_001) and the differences between the two products are reported: 90-95% of ice edge locations are the same.

In case of differences, experience shows that it is more frequent that the ice chart edge shows more ice than the OSI SAF concentration product. Between 80 and 85% of grid points have differences less than 10% and more than 90% of points have differences less than 20%.

Standard deviation of differences between the ice charts and the OSI SAF concentration are between 7% and 9% for the analysed period. Biases are in the order of 5%, with the ice chart overestimating relative to the OSI SAF product.

A report on estimating the ice chart uncertainty from comparison analyses of different ice chart data sets is included as an Appendix to this document.

I.2.2 Antarctic Ocean – High Resolution Sea Ice Information

SEAICE_ANT_PHY_AUTO_L3_NRT_011_012

The DMI-ASIP sea ice classification dataset (cmems_obs-si_ant_phy_nrt_1 km_P1D) is produced by DMI with the ASIP (Automated Sea Ice Products) deep learning model. The products are

[3] See https://www.asi.it/en/earth-science/cosmo-skymed/ (last accessed on 01/11/2025)

automatically retrieved from Sentinel-1 SAR satellite images (Extra Wide -EW- and Interferometric Wide -IW- swath mode) combined with passive microwave data from the Global Change Observation Mission – Water (GCOM-W) Advanced Scanning Microwave Radiometer 2 (AMSR2) satellite sensor[4]. The automated interpretation of sea ice and water signatures in the two satellite data sources is done using a convolutional neural network (CNN) that has been trained with DMI regional ice charts, using the ASIP/AI4Arctic Sea Ice Training Dataset[5]. All Sentinel-1 scenes that are processed within a day (in UTC) are collected into a daily mosaic where the latest acquired satellite imagery is overlaid the older imagery, and with information of satellite acquisition time given for each grid cell.

The dataset covers Antarctic waters and has a spatial horizontal resolution of 1 x 1 km. The DMI-ASIP automated ice concentration data gives sea ice concentration for every 10% (0% (open water); 10%; 20%; ...; 100%) in each grid cell.

The DMI-ASIP sea ice data are compared to 24 selected test data DMI regional ice charts for the Antarctic. The ice charts are matched with the relevant Sentinel-1 imagery and gridded onto the S-1 geometry. The ice chart sea ice concentration (SIC) and DMI-ASIP SIC are compared on grid scale: for every grid point where the ice chart SIC is e.g., 10%, the corresponding DMI-ASIP SIC value is subtracted from the ice chart SIC. The mean difference of DMI-ASIP SIC to all 10% ice chart grid points is calculated together with the standard deviation. The same is done for ice chart SICs of 0%, 20%, 30%, up to 100%.

The results of the comparison show good agreement between the two products where mean and std. of the DMI-ASIP product for ice chart SIC values of 0%, 50% and 100% are 0.7 +/-6.2, 59.9 +/-22.1 and 99.1 +/-5.4 respectively. The DMI-ASIP data files also contain information about the model confidence in the SIC predictions and a validation of these shows that the model confidence estimate is reliable.

I.3 Estimated Accuracy Numbers

This section includes the name of the products and datasets, the units and a description of the metrics as well as the values for those metrics (see Table 2 below).

[4] See https://suzaku.eorc.jaxa.jp/GCOM_W/w_amsr2/whats_amsr2.html (last accessed on 01/11/2025)

[5] ASIDv2 training data: https://doi.org/10.11583/DTU.13011134.v3 (last assigned 12-11-2024)

Table 2: Summary of the Estimated Accuracy Numbers in terms of mean, bias and/or standard deviations derived from assessments of each of the datasets covered in this document against reference datasets

Product Name	Dataset	Metric description[6]	Accuracy Numbers
SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_002	Ice concentration, Svalbard ice chart	Only a visual validation against optical satellite data from Sentinel-2, NOAA AVHRR and MODIS.	Visual validation: Generally a good agreement.
			Visual validation: Generally a good agreement.
	Ice concentration, Greenland ice charts	Mean bias and standard deviation (STD) of the bias, from comparison against OSI SAF ice concentration.	Bias ± STD for OSISAF-SIC 0-100%: ~5% ± 7-9%
SEAICE_ANT_PHY_AUTO_L3_NRT_011_012	Ice concentration, Antarctica product	Mean bias and standard deviation (STD) of the bias, when compared against ice chart SIC (DMI ice charts for the Antarctic Ocean).	Bias ± STD for DMI-SIC 0%: 0.7 ± 6.2% Bias ± STD for DMI-SIC 50%: 59.9 ± 22.1% Bias ± STD for DMI-SIC 100%: 99.1 ± 5.4%

[6] Typically, in validation of sea ice concentration compared with ice charts, one refers to the standard deviation and bias. For each ice concentration level in the ice chart, the deviation between ice chart concentration and OSI SAF ice concentration is calculated. Later deviations are grouped into categories, i.e. +/-10% and +/-20%. Furthermore, the bias and standard deviation is calculated for each concentration level. The bias and standard deviation are reported for ice (> 0% ice concentration), for water (0% ice concentration) and for both ice and water as a total.

II PRODUCTION SYSTEM DESCRIPTION

The Sea Ice Thematic Assembly Center (SI TAC) is a multi-mission integration activity. The main tasks to fulfil are:

1. provide real-time, delayed mode update operations for L3, and L4 data products for Sea Ice;
2. provide data for long-term sea ice monitoring data (climatology);
3. provide quality control, validation and error characterization of data products and services.

Several Production Units process and deliver sea ice products and perform the routine quality monitoring and validation of their products.

For several of the sea ice products that are delivered, satellite data from SAR are used as the main source of input data due to its high spatial resolution and independence from the weather. During the MyOcean projects[7] (precursor to CMEMS from 2009 to 2015) the sea ice partners established a very good working relationship with the European Space Agency (ESA) in order to ensure access to SAR data, and this cooperation continues in CMEMS. In the preparation of the Sentinel-1a mission there were discussions regarding the geographical coverage and requested beam modes. Through these discussions the partners were able to ensure the best possible access for their services. A direct link to the ESA datahub was established and data from Sentinel-1 are retrieved in NRT by automatically polling for new data files on the ftp server. After a new file has been detected it is transmitted and processed at the local production unit. The process is similar as for RADARSAT.

II.1 Arctic Ocean – Sea Ice Concentration Charts – Svalbard and Greenland

SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_002

The sea ice concentration ice charts from the national Ice service are generated based on a manual interpretation of high-resolution satellite data. Satellite data from Sentinel-1 and Radarsat-2 are automatically downloaded from the ftp servers at ESA-data hub. Sentinel-1 is provided in Extended Wide mode dual polarization (HH/HV). One image covers 400x400 km and has a spatial resolution of 20x40 meters. Radarsat-2 data are provided in ScanSAR Wide mode dual polarization (HH/HV). One image covers 500x500 km and has a spatial horizontal resolution of 50x50 meter. In addition to these SAR data, the analyses are supplemented with optical satellite data from Sentinel-2, Sentinel-3, the Moderate Resolution Imaging Spectroradiometer (MODIS) and NOAA Advanced Very High Resolution Radiometer (AVHRR). The data are made available for the ice analyst/operator in NRT and are manually analysed with respect to sea ice concentration in a Geographic Information System (GIS) production system. The Baltic Sea ice charts also provide the average ice thickness, and this parameter is also provided as one SEAICE_BAL_SEAICE_L4_NRT_OBSERVATIONS_011_004 dataset. Also, the sea ice extent (SIE)

[7] MyOcean project granted by the European Commission within the GMES Program: https://www.copernicus.eu/en/myocean (last accessed 12-11-2024)

derived from the SIC data over the Baltic Sea set is provided as a separate dataset. The final analyses are gridded to a 1 km NetCDF product.

II.2 Antarctic Ocean - High Resolution Sea Ice Information

SEAICE_ANT_PHY_AUTO_L3_NRT_011_012

The DMI-ASIP sea ice classification - Antarctic (cmems_obs-si_ant_phy_nrt_1km_P1D) dataset is produced by DMI. The products are automatically retrieved from radar (SAR) satellite images (Extra Wide swath and Interferometric Wide swath mode) with ~80 m pixel spacing (both image modes are resampled onto a 80 m grid) from the Copernicus Sentinel-1 satellite, combined with passive microwave data from the AMSR2 sensor onboard the Japanese GCOM-W satellite. The automated interpretation of sea ice and water signatures in the two satellite data sources is done using a convolutional neural network (CNN) that has been trained with DMI regional ice charts, using the ASIP/AI4Arctic Sea Ice Training Dataset (ASIDv2[8]). All Sentinel-1 scenes that are processed within a day (0:00 – 23:59 UTC) are collected into a daily mosaic (newest information on top) with information of satellite acquisition time given for each grid cell. The product covers the Antarctic Ocean and has a spatial horizontal resolution of 1 x 1 km.

[8] Training dataset: [8], https://doi.org/10.11583/DTU.13011134.v3 (last accessed 12-11-2024)

III VALIDATION FRAMEWORK

III.1 Arctic Ocean – Sea Ice Concentration Charts – Svalbard and Greenland

This section provides a description of the validation used for the ice charts from the national ice services at the Norwegian and Danish meteorological institutes. Given the absence of ground truth data for validating satellite-derived sea ice parameters, our validation methods primarily focus on intercomparing of satellite-based products. The high-resolution ice charts, generated by the national Sea Ice Services, are based on manual interpretation of all available satellite data, which include data from optical, SAR, and passive microwave sensors. The considerable number of input sources, the high spatial resolution, and the manual control of the output often lead to these ice charts being used as "Ground Truth." There are however, some subjectivities in the analyses that are challenging to quantify. The Copernicus product name and info are given in **Error! Reference source not found.**.

Table 3: SI TAC, product name, description of the product, production unit, and the production center. The datasets are produced by the national ice service at MET-Norway and the Danish Met Institute, covering the Svalbard and the Greenland area, respectively.

Product	*Product description*	*Production unit (PU)*	*Production Centre (PC)*
SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_002	Arctic sea ice L4 – Svalbard, Greenland	SIW-METNO-OSLO-NO, SIW-DMI-COPENHAGEN-DK	SIW-METNO-OSLO-NO

III.1.1 Svalbard ice charts

Dataset in SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_002

The regional sea ice product Svalbard is provided by the Sea Ice Service in Tromsø, VNN (Værvarslinga for Nord-Norge, MET). It covers European Arctic with focus on the areas around Svalbard and the Barents Sea. The ice concentration is given with 1 km horizontal resolution. The ice charts are primarily based on SAR (Radarsat-2) data, together with AVHRR and MODIS data. A detailed interpretation of satellite imagery and a subsequent mapping procedure are carried out by skilled ice analysts.

An important use of the products is for validation of ocean/ice models (Arctic and Global Monitoring and Forecasting Center -MFC-) and the global SI TAC products.

Validation of the products is a challenge due to lack of ground truth. The routine comparison with OSI SAF global ice concentration and ice edge data allows for more regular data quality assurance. The product is at irregular intervals manual compared against medium-resolution optical satellite data from the MODIS and NOAA satellites. The validation is focused on situations where weather conditions allow the use of optical data. In addition, some alternative SAR data

from the COSMO-SkyMed satellites have been used for validation. Validation is not carried out in the dark session due to lack of daylight and no visual data available.

III.1.2 Greenland ice chart

Dataset in SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_002

The regional Sea Ice products Greenland cover Greenland Waters and are produced and delivered by DMI. These are high-resolution ice concentration with grid resolution of 1 km.

The product comes in two variants. The regional 'NRT' ice chart which typically covers an approximately 500 x 500 km subset of the area as observed by a radar satellite, and the 'overview' ice chart which is produced twice per week and covers the entire area. The nominal quality of the charts is similar, and validation procedures are the same.

The products (ice charts) are based on manual interpretation of high-resolution satellite data, primarily SAR imagery from Sentinel-1, RADARSAT, TerraSAR-X, Cosmo Skymed, MODIS, NOAA- and METOP-AVHRR, AMSR2 and SSMIS (Special Sensor Microwave Imager / Sounder).

Ice charts are originally produced as vector charts using a GIS software and a large number of satellite data sources. The vector charts (polygons of areas with similar ice conditions) are subsequently gridded on a 1 km grid and converted to NetCDF for CMEMS distribution and use. Boundaries of polygons are drawn with a nominal accuracy of approximately 1 km.

The main use of the data in CMEMS is for validation of ocean/ice models (Arctic and Global MFC) and the global Sea Ice Thematic Assembly Centre (SI TAC) products.

Validation of the high-resolution regional ice chart of the Arctic Ocean/Greenland Waters is a challenging task due to lack of ground truth. In the production, the satellite data used is expected to have a resolution high enough to represent the ground truth of the mapped area. The interpretation of the satellite data is a subjective analysis by the operator on duty. To verify how this influenced the final product, two different ice analysts produce independent products based on the same input data at regular intervals. The two products will be compared on a pixel basis and a confusion matrix containing ice concentration classes from the two analysts will be generated (V4). This will give a measure of the uncertainty of the different ice concentration classes and the overall uncertainty of the product. We also continuously compare the regional charts with the OSI SAF global sea ice analysis.

III.2 Antarctic Ocean - High Resolution Sea Ice Information

The Antarctic high resolution sea ice product is produced by DMI based on an automatic deep learning model trained for classify sea ice concentration based on satellite data from SAR and AMSR2. The product has been validated against the manually derived ice chart from the Danish ice service. Table 4 shows the Copernicus product name and info of the product.

Table 4: SI TAC, product name, description of the product, production unit, and the production centre. The datasets are produced by the Danish meteorological institute.

Product	*Product description*	*Production unit (PU)*	*Production Centre (PC)*
SEAICE_ANT_PHY_AUTO_L3_NRT_011_012	Antarctic high-resolution ice product	SIW-DMI-COPENHAGEN-DK	SIW-METNO-OSLO-NO

The DMI-ASIP sea ice classification dataset is produced by DMI using the ASIP (Automated Sea Ice Products) deep learning model. The products are automatically retrieved from radar (SAR) satellite images (Extra Wide swath and Interferometric Wide swath mode) with ~80 m pixel spacing (both image modes are resampled onto an 80 m grid) from the Copernicus Sentinel-1 satellite, combined with passive microwave data from the AMSR2 sensor onboard the Japanese GCOM-W satellite. The automated interpretation of sea ice and water signatures in the two satellite data sources is done using a convolutional neural network (CNN) that has been trained with DMI regional ice charts, using the ASIP/AI4Arctic Sea Ice Training Dataset[12]. See examples of an input Sentinel-1 dual-pol (HH and HV) image and the corresponding DMI ice chart and DMI-ASIP output ice map in Figure 3. All Sentinel-1 scenes that are processed within a day (in UTC) are collected into a daily mosaic (newest information on top) with information of satellite acquisition time given for each grid cell.

The dataset covers the Antarctic Ocean and has a spatial horizontal resolution of 1.0 x 1.0 km. The DMI-ASIP automated ice concentration data gives sea ice concentration for every 10% (0% = open waters; 10%; 20%; ...; 100%) in each grid cell.

The automatically generated sea ice data is compared with the DMI Ice Service manually produced regional ice chart products for the Antarctic Ocean that is used as the reference dataset. The ice charts are a detailed interpretation of primarily SAR satellite imagery and a subsequent mapping procedure is carried out by skilled (experienced and trained) ice analysts drawing polygons of sea ice concentration (SIC) areas.

[12] ASIDv2 dataset used for training: https://doi.org/10.11583/DTU.13011134.v3 (latest available 13-11-2024)

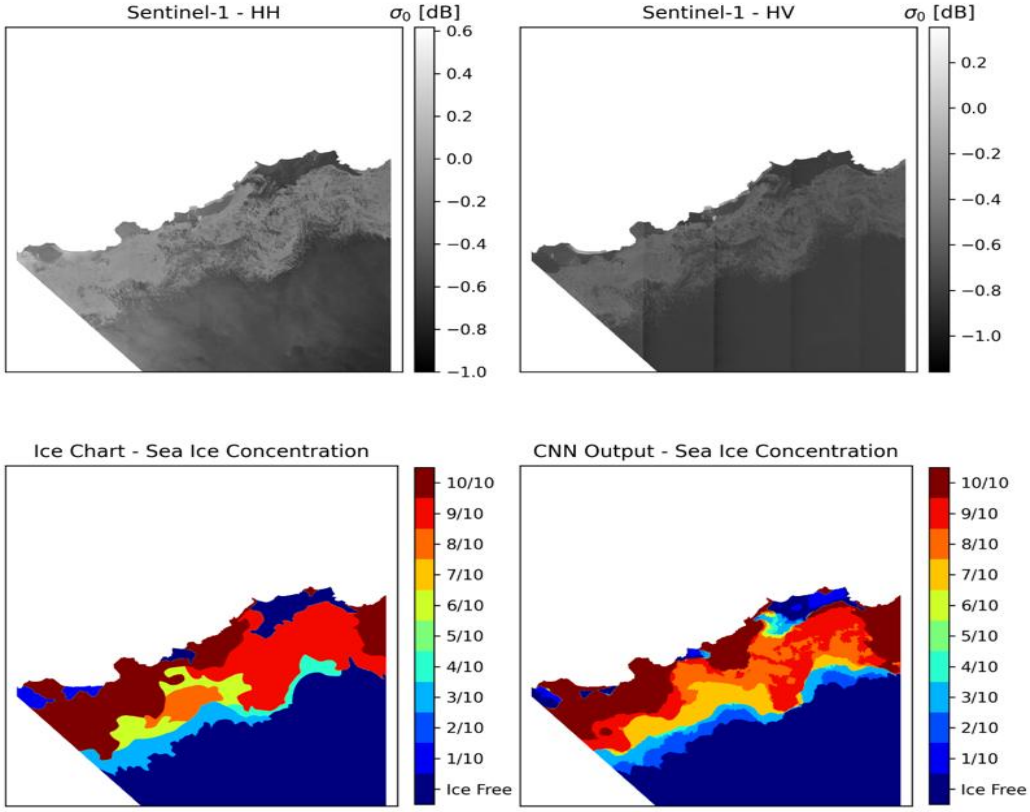


Figure 3: Example of DMI-ASIP data file (bottom right) and the reference DMI regional ice chart (bottom left) and the Sentinel-1 image HH (top left) and HV (top right) channels that are used as input to the DMI-ASIP classification algorithm with AMSR2 data.

The ice chart classifies concentration according to the World Meteorological Organization (WMO) standard for every 10% ice concentration (0% -open water-, 10%, 20%, up to 100%) or in intervals (e.g. 10-30%). The DMI regional ice charts for the Antarctic Ocean are produced only in December and January, but in that period, they are produced almost daily (with irregular frequency, depending on the acquisition of relevant SAR imagery) and are primarily used for strategic and tactical planning within the offshore and shipping community. The DMI regional ice charts are matched with Sentinel-1 imagery and gridded onto the Sentinel-1 geometry. We have selected 24 ice charts from the dataset for this verification. The ice chart SIC and DMI-ASIP SIC are compared on grid scale: for every grid point where the ice chart SIC is (e.g., 10%, the corresponding DMI-ASIP SIC value is subtracted from the ice chart SIC). The mean difference of DMI-ASIP SIC to all 10% ice chart grid points is calculated together with the standard deviation. The same is done for ice chart SICs of 0%, 20%, 30%, up to 100%. The resulting statistics are shown below.

IV VALIDATION RESULTS

IV.1 Arctic Ocean – Sea Ice Concentration Charts – Svalbard and Greenland

This section provides a description of the validation used for the ice charts from the national ice services at the Norwegian and Danish meteorological institutes. Given the absence of ground truth data for validating satellite-derived sea ice parameters, our validation methods primarily focus on intercomparing of satellite-based products. The high-resolution ice charts, generated by the national Sea Ice Services, are based on manual interpretation of all available satellite data, which include data from optical, SAR, and passive microwave sensors. The considerable number of input sources, the high spatial resolution, and the manual control of the output often lead to these ice charts being used as "Ground Truth." There are however, some subjectivities in the analyses that are challenging to quantify. The Copernicus product name and info are given in **Error! Reference source not found.**.

IV.1.1 Svalbard ice chart

Dataset in SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_002

The validation of the regional high-resolution ice chart of the Arctic has been performed in a two-step routine. First the accuracy of the product has been validated against alternative satellite sources. Secondly, the produced NetCDF file has been checked to be compliant with the CF-1.4 version of the NetCDF standard[13].

Figure 4 shows an example of the routine intercomparison against the Ocean and Sea Ice Satellite Application Facility (OSI SAF) sea ice concentration products[14]. For each ice chart concentration level the deviation between ice chart concentration and OSI SAF ice concentration is calculated. Afterwards deviations are grouped into categories, i.e. ±10% and ±20%. Furthermore the mean difference and standard deviation are calculated and reported for ice (100% ice concentration) and for water (0% ice concentration). Conventional mean difference and standard deviations are used for all calculations.

[13] CF Conformance Requirements and Recommendations 1.4: https://cfconventions.org/Data/cf-documents/requirements-recommendations/requirements-recommendations-1.4.html (last accessed 06-10-2024).

[14] See OSI SAF Half-Year Validation report: https://osi-saf.eumetsat.int/sites/default/files/documents/documents/file/osisaf_cdop4_rp_hyr25_h1_1.0.pdf (last accessed 01/11/2025)

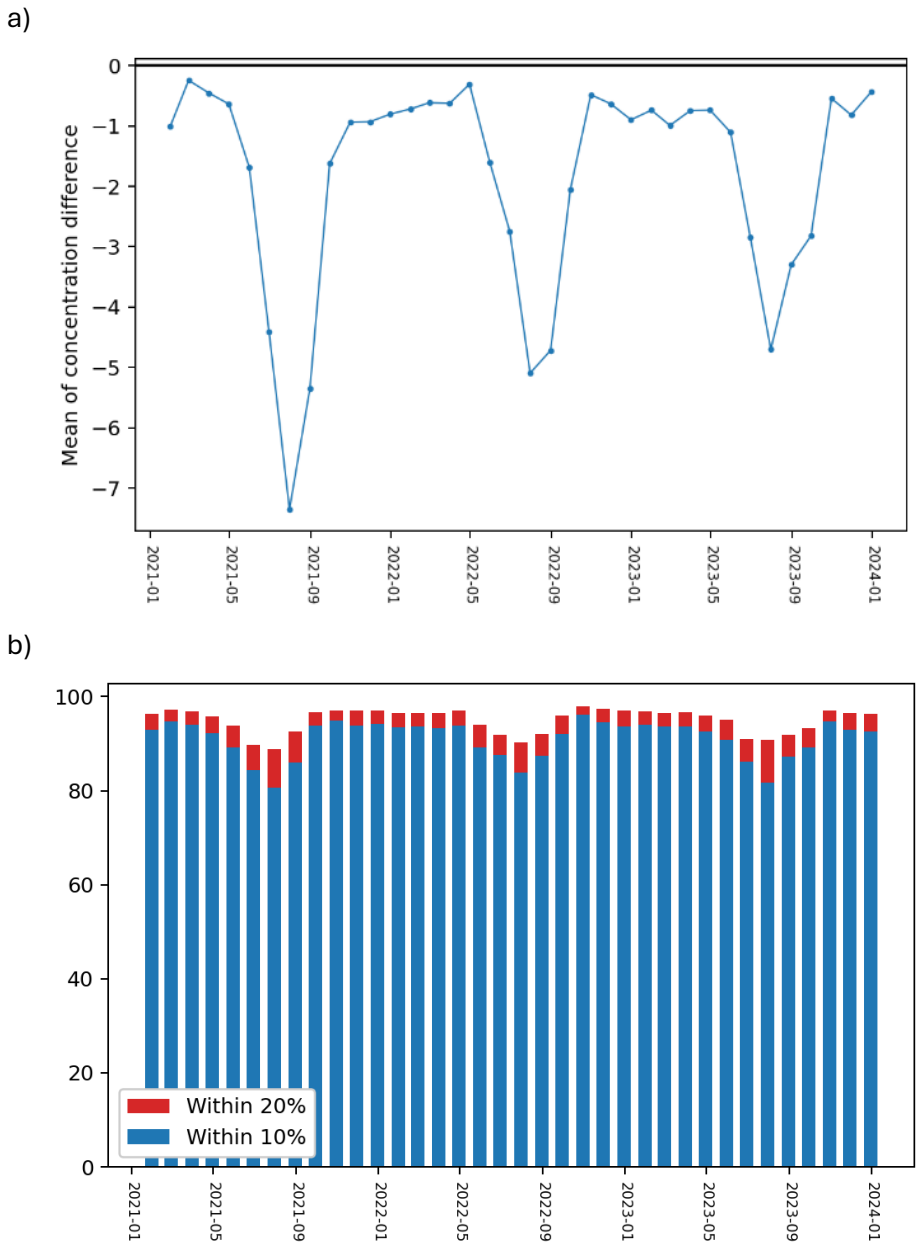


Figure 4: Results from intercomparing ice chart sea ice concentration against the OSISAF (OSI-401-d) sea ice concentration, (a) mean SIC difference through 2021 to 2023 and (b) total grid cells percentual match in SIC within 20% and 10% of the reference data SIC.

Figure 5 shows an example of the intercomparison of ice charts versus the Arctic OSI SAF ice edge product.

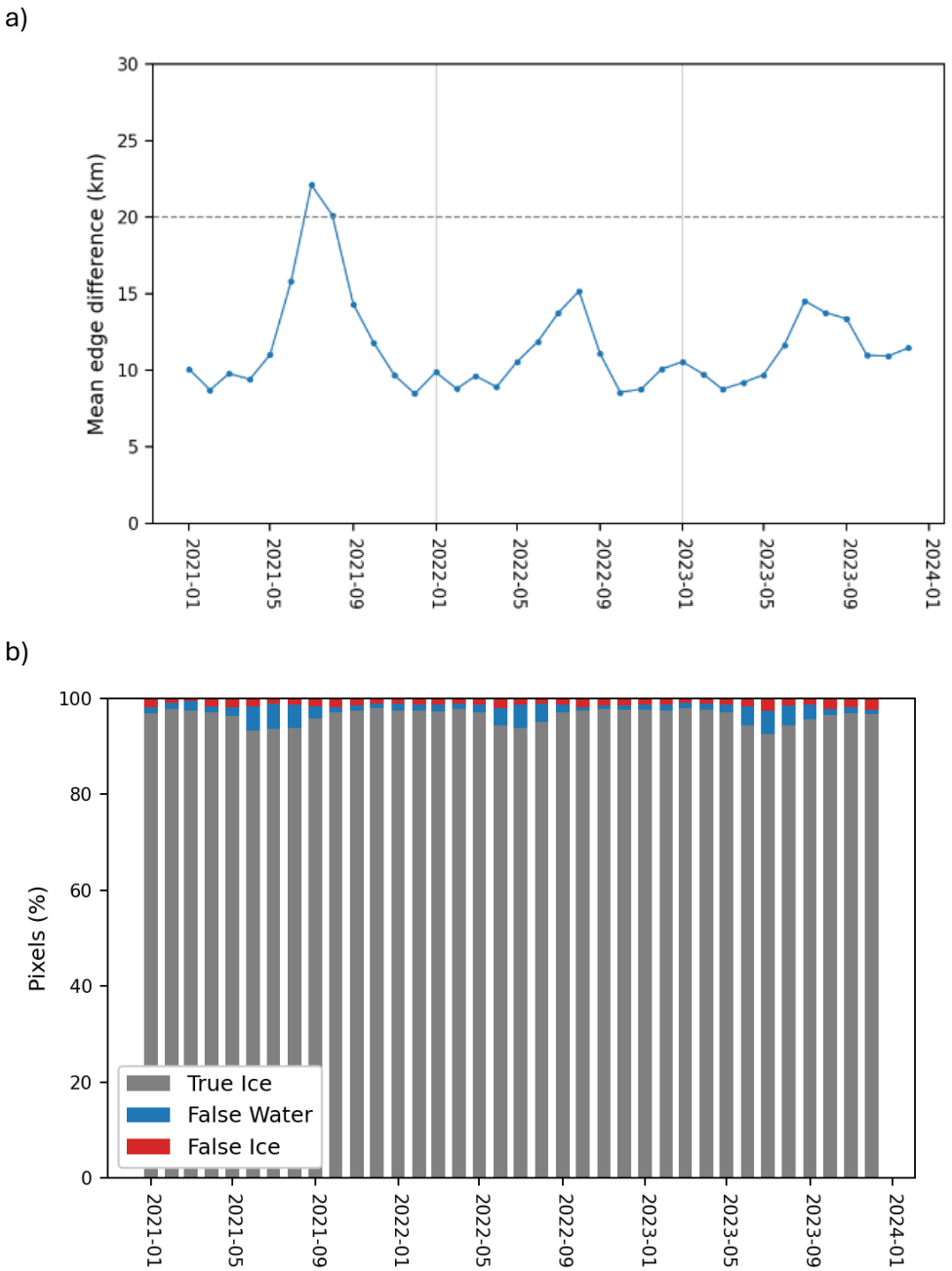


Figure 5: Results from comparing the ice chart sea ice concentration verses the OSISAF (OSI-402-d) ice edge product; (a) mean edge difference between the two products, while the (b) shows matching true ice, false ice and false water.

A more detailed description of the comparison between ice chart and OSISAF products can be found in Figure 6, which shows an example of a visual validation against a NOAA image. Figure 7 shows visual validation against a COSMO-SkyMed image[15].

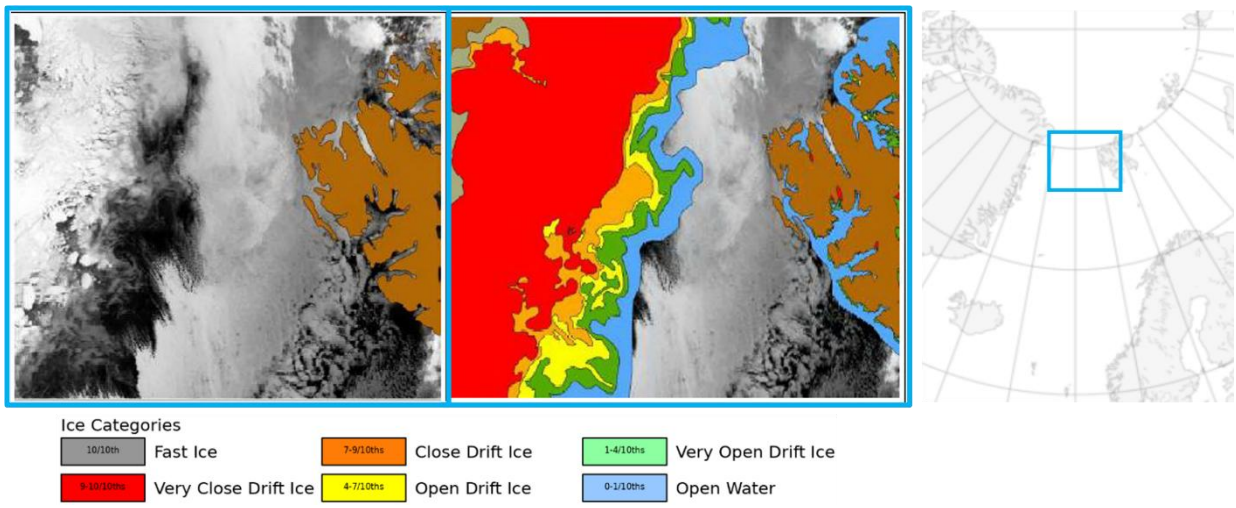


Figure 6: Visual validation of the regional high-resolution ice chart (right) against a NOAA AVHRR image, northwest of Svalbard (left), 8 February 2013. The lower part shows the legend of the ice charts concentration classes.

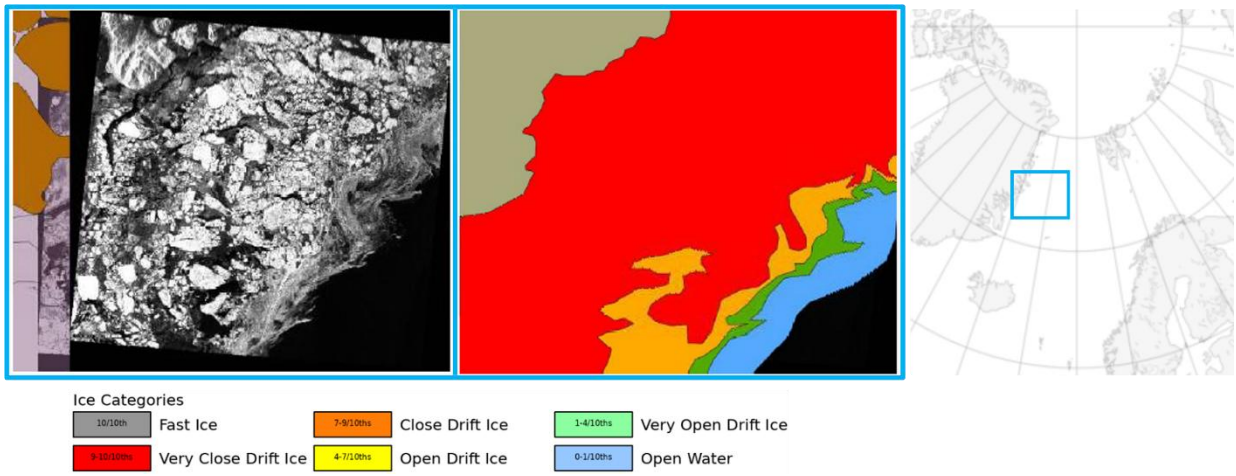


Figure 7: Visual comparison of COSMO-SkyMed data (left) and ice chart (right) on 10 December 2012.

[15] Latest validation reports; https://osisaf-hl.met.no/sea-ice-conc-edge-validation (last accessed 12-11-2024).

IV.1.2 Greenland ice charts

Dataset in SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_002

Since we do not know whether OSI SAF or the ice charts in general are most correct, we report only on the differences between the two. Figure 8 show the validation results for the 1st quarter of 2013. The latest validation report[16] and the Validation of the DMI overview charts and the DMI regional charts give the same results.

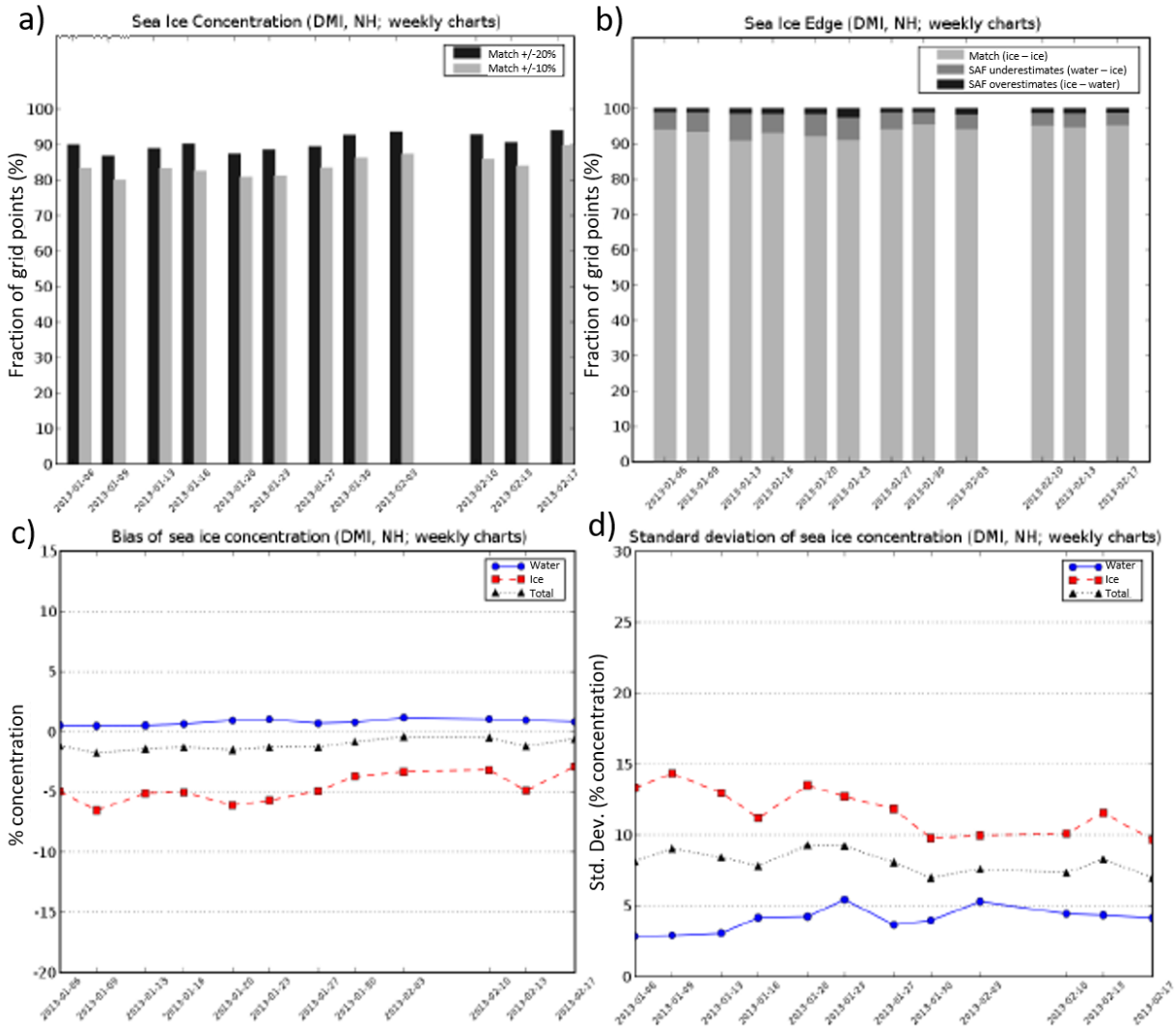
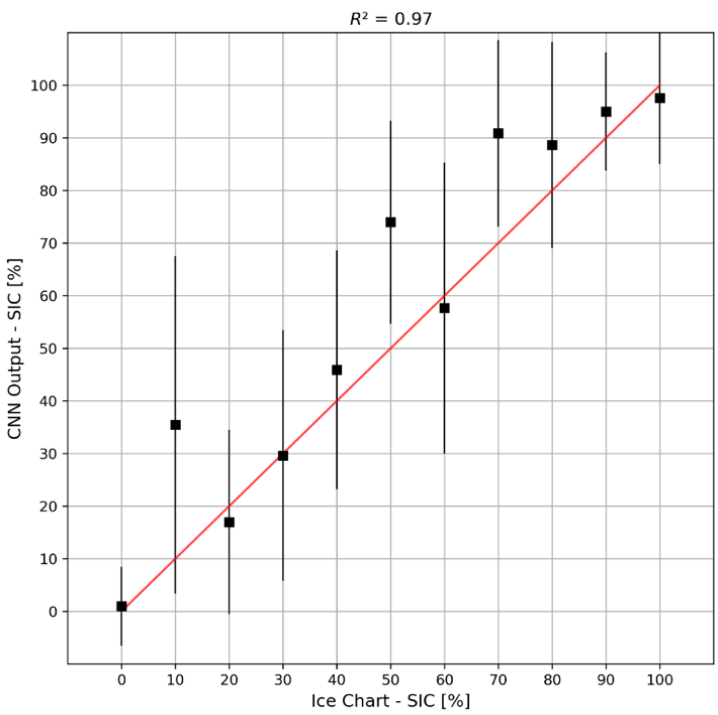


Figure 8: Comparison between OSI SAF ice concentration (SEAICE_GLO_SEAICE_L4_NRT_OBSERVATIONS _011_ 001, ice edge dataset) and DMI Greenland ice charts (SEAICE_ARC_SEAICE_ L4_NRT_OBSERVATIONS_011 _002). Assessment results of the comparison between OSI SAF and DMI ice charts, as the a) match of ice concentrations, b) match of ice edges locations, c) bias and d) standard deviation of the bias between the OSI SAF and DMI ice charts ice concentrations.

[16] Latest validation report: http://myocean.met.no/SIW-TAC/siwdocs.html (last accessed 12-11-2024)

IV.2 Antarctic Ocean - High Resolution Sea Ice Information

The results from the comparison of DMI-ASIP (CNN-based) automated sea ice data against a test dataset of 24 selected DMI regional ice charts are shown in the plot and Table in as the mean ice concentration and standard deviation for each 10% ice concentration value. The DMI-ASIP data files also contain information about the model confidence in the SIC predictions. The reliability diagram in shows that the model confidence estimate is reliable. The assessment of the model reliability is based on an independent validation dataset. We calibrate our model using a post-processing technique known as temperature scaling (Guo et al., 2017) to improve the model´s reliability.



ASIP prediction		
SIC	*mean*	*std*
0	0.7	6.2
10	10.3	19.1
20	19.6	20.7
30	29.9	24.3
40	42.6	23.4
50	59.9	22.1
60	57.5	23.3
70	68.4	26.6
80	75.9	22.3
90	90.1	15.5
100	99.1	5.4

Figure 9: Plot of the comparison of DMI-ASIP (CNN-based) automated sea ice concentration products against 24 selected DMI regional ice charts ice concentrations, for every 10% ice concentration: DMI-ASIP mean values (black squares) and standard deviation (vertical black lines) for each ice chart concentration value. x=y is shown as a red line as a reference. The coefficient of determination R^2 , being the square of the Pearson correlation coefficient, R, quantifies the degree of linear correlation between DMI-ASIP and Ice chart ice concentrations is provided just above the plot. Mean values and std.dev. are also given in the table to the right of the plot.

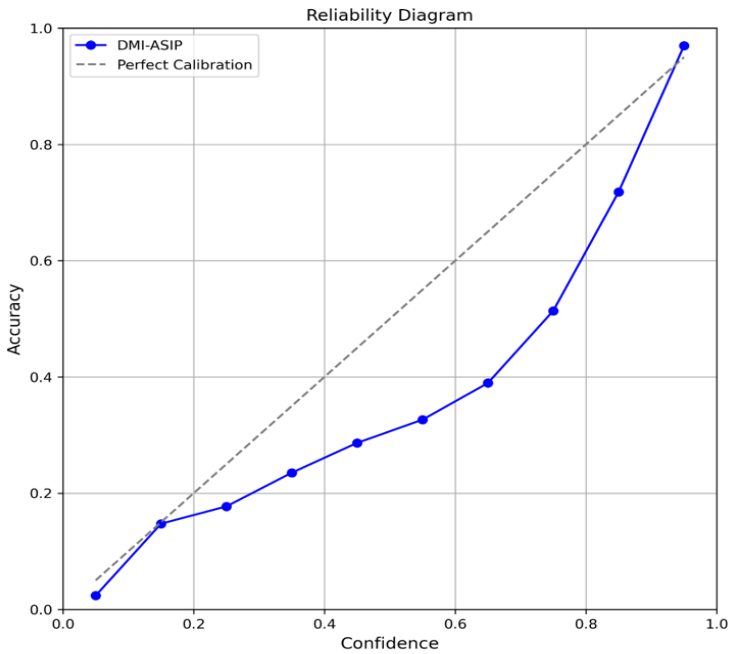


Figure 10: Reliability diagram showing the bin-wise predictive accuracy as a function of the bin-wise predictive confidence for the DMI-ASIP model evaluated against a test dataset of 24 selected regional Antarctic ice charts produced by the Greenland Ice Service at DMI. The predictive accuracy and the predictive confidence are computed for 10 confidence bins (0-5%, 5-15%, 15-25%, ..., 95-100%). Perfect calibration is indicated by the identity function, and any deviation from the diagonal represents miscalibration.

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

This section is dedicated to describe and track changes that may occur in the current operational system. It includes changes due to outages (data gaps), upstream data (e.g., addition or loss of a sensor, inclusion/removal of in-situ platforms), reprocessing, corrections, assimilated data, boundary conditions, among others, that may impact product quality (see Table 6 below).

Table 5: List of noticeable events, outages or changes

Date	Change/Event description	System version	other
	No changes.		

VI QUALITY CHANGES SINCE PREVIOUS VERSION

This section aims to provide a short summary of the main changes in Product Quality between the previous and current version. Nothing to report in this version as no quality change has taken place since previous version.

VII REFERENCES

Guo, C., Pleiss, G., Sun, Y., & Weinberger, K. Q. (2017, July). On calibration of modern neural networks. In *International conference on machine learning* (pp. 1321-1330). PMLR.

VIII APPENDIX

VIII.1 Ice Chart Uncertainty Estimates

This section describes a study carried out during the MyOcean-2[17] project by the ice charting Production Units of the MyOcean OSI-TAC. The same organizations are now delivering to the CMEMS SI TAC service, so the results are fully applicable to the current CMEMS products of the same names.

High resolution ice charts produced by the national Sea Ice Services of Denmark, Finland and Norway are in MyOcean made available to MFCs in a NetCDF format that are usable for validation of models and sea ice data assimilation. Uncertainty estimates are lacking in practically all fundamental ice chart data sets available. Therefore, a main task in MyOcean-2 has been to investigate the possibility for development of common / standardized validation methods for the high resolution sea ice analysis, with the aim of enabling the computation of consistent and comparable ice chart uncertainty estimates and error statistics, in terms of possible biases, variance and true resolution of the ice charts. Since there is not yet a better automatic tool for interpretation of sea ice in high resolution imagery, and due to the lack of field observations, ice charts are considered being the data sets that best describes the true state of the sea ice, and thus other sea ice data sets uses these as reference.

The intercomparison of ice chart data sets and the analysis of differences is a useful way to evaluate the realism of the data sets and thus some estimates of ice chart uncertainty. This study seeks to do a systematic and comprehensive cross comparison of existing ice chart products from multiple MyOcean ice chart data sets over the same areas, as well as a comparison with the OSI SAF ice concentration (IC) data set (also available through MyOcean), to characterize important temporal and spatial differences in the data sets. First approach has been to compare the ice chart IC data on full scale; all data, all common areas. Then, look at temporal and spatial differences; winter/summer data, individual years of data, local area comparisons.

DMI has done comparison analysis of DMI Greenland Ice charts and MET Svalbard Ice charts (SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_002), both covering Svalbard and Greenland waters. An additional comparison of the DMI and MET Ice charts with OSI SAF IC data set (SEAICE_GLO_SEAICE_L4_NRT_OBSERVATIONS_011_001) was made for reference. This work has been done with input and in constructive discussion with MET. Concurrently, FMI has done a similar comparison of FMI Ice charts (SEAICE_BAL_SEAICE_L4_NRT_OBSERVATIONS_011_004) and Swedish Meteorological and Hydrological Institute (SHMI) Ice charts both covering the Baltic Sea. The main results from these comparison analyses are presented in the QUID associated to SEAICE_BAL_SEAICE_L4_NRT_OBSERVATIONS_011_004.

[17] MyOcean project granted by the European Commission within the GMES Program: https://www.copernicus.eu/en/myocean (last accessed 12-11-2024)

VIII.1.1 Comparison of Arctic ice charts and OSI SAF ice concentration dataset

Danish Meteorological Institute (DMI) and Norwegian Meteorological Institute (MET) both publish ice charts covering Svalbard and Greenland waters (SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_ 011_002). DMI produces an Overview Ice chart on Mondays and Thursday and Regional Ice charts covering different sub-areas on an irregular basis. MET produces a Svalbard Ice chart on every weekday.

Through the MyOcean Online Data Catalogue, both the DMI and MET ice chart IC data sets are available in NetCDF format in a 1 km grid (nominal resolution of the ice charts). The comparisons were made for the common area of the DMI and MET ice charts for the whole season. Additionally, the same comparison analysis was made between the DMI and MET ice chart data sets and the OSI SAF IC data set (SEAICE_GLO_SEAICE_L4_NRT_OBSERVATIONS_011_001) that covers the whole Arctic. This data set is also available through the MyOcean Online Data Catalogue in NetCDF format in a 10 km grid. Comparisons were also made on seasonal basis (summer, winter) and on separate years. Also, a local comparison in the Northeast Greenland area was made. Only the results of the comparison of the DMI ice charts, the MET ice charts and the OSI SAF IC dataset for the full season are shown here. The parameter compared in this study was ice concentration. The data time series used in the comparison are what was available in the catalogue up till November 2013. Examples of the four sea ice concentration products targeted in the comparison exercise are shown in Figure 11, 12 and 13 below.

For each data set comparison, the approach has been to find the data samples that are produced on the same day and to compare ice concentration values (and also ice thickness values for the Baltic ice charts) on grid cell level. Table 7 gives general information on the dataset intersection in terms of time series and number of products and ice charts with a timely and geographical overlap. The data sets have different characteristics that are described in the corresponding Product Users Manuals (PUMs). There are two sets of results for each data set comparison, where each of the datasets serves as the basis for the comparison. Results are plotted in a blurred scatter plots in Figure 14 to 22 that visualizes the frequency of the data values; white numbers are the occurrence of data values in (rounded) percent (of the total number of corresponding ice concentration values). Zero indicates an occurrence of less than half a percent. The Tables 8 to 15 show statistics – mean, median and standard deviation – for each unique ice concentration in the dataset on the value axis. The large number of corresponding open water (0% ice concentration) grid cells in the data sets has been neglected in the comparisons, so that only corresponding grid cells where there is ice in one of the data sets, is taking into account. If the many corresponding open water grid cells were included in the analysis, the results for 0% ice concentration would undoubtedly have a mean of 0% and an insignificant standard deviation.

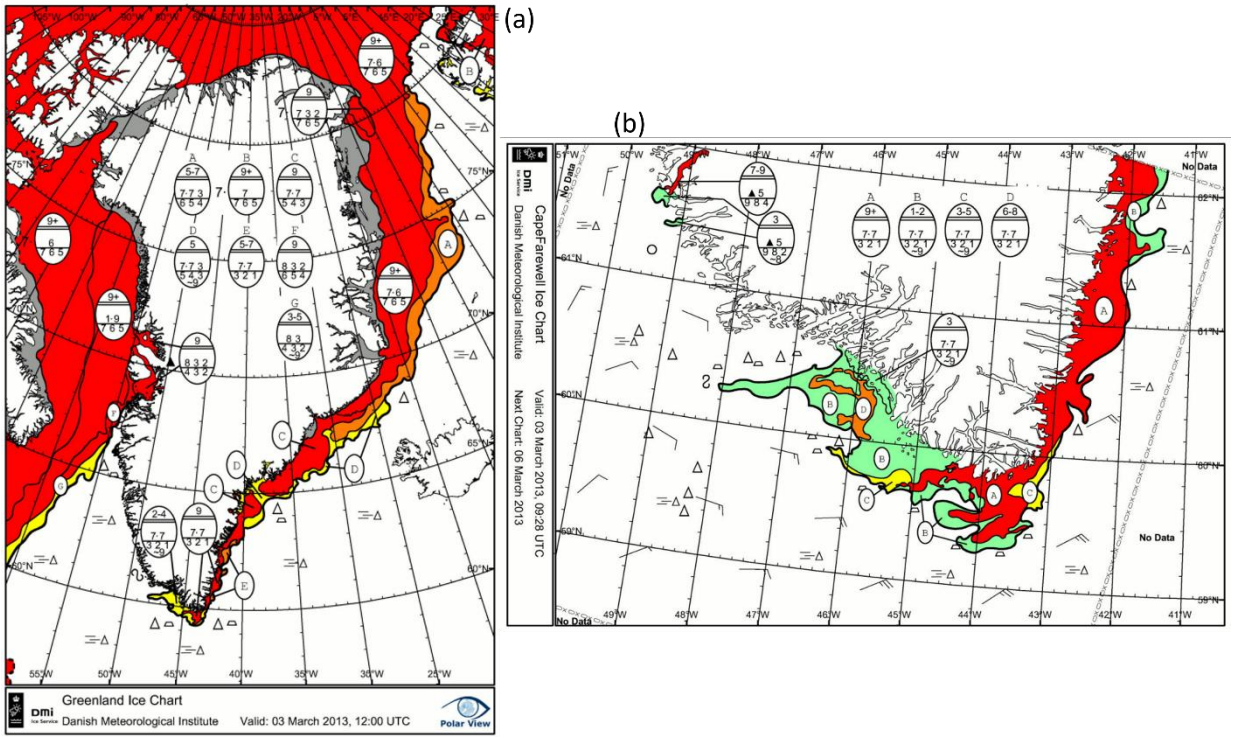


Figure 11: Examples of the two variants of the DMI Ice Service ice charts, (a) a Greenland Overview Ice chart produced for March 3rd 2013 and (b) a regional ice chart of Southern Greenland produced for March 2nd 2013.

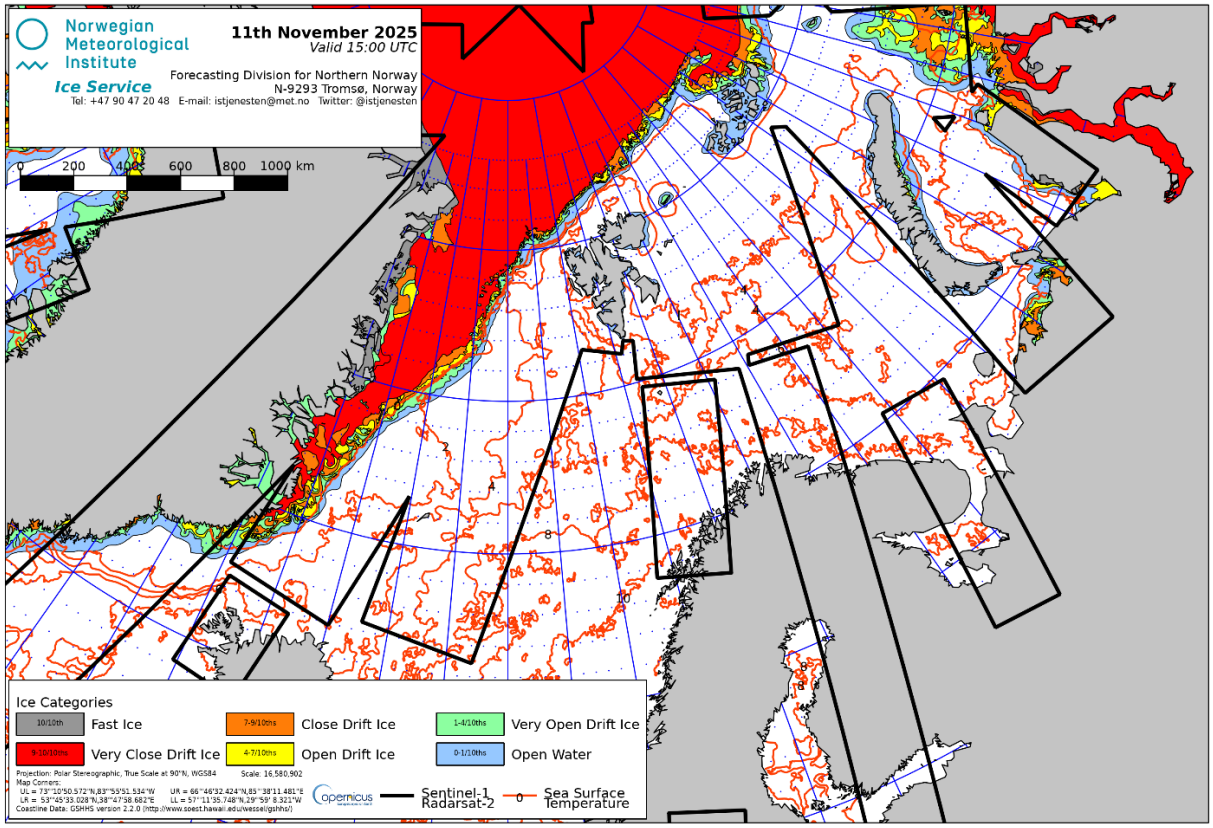


Figure 12: Example of the MET Ice Service ice chart of the Arctic North Atlantic region and Svalbard produced for November 11th 2025.

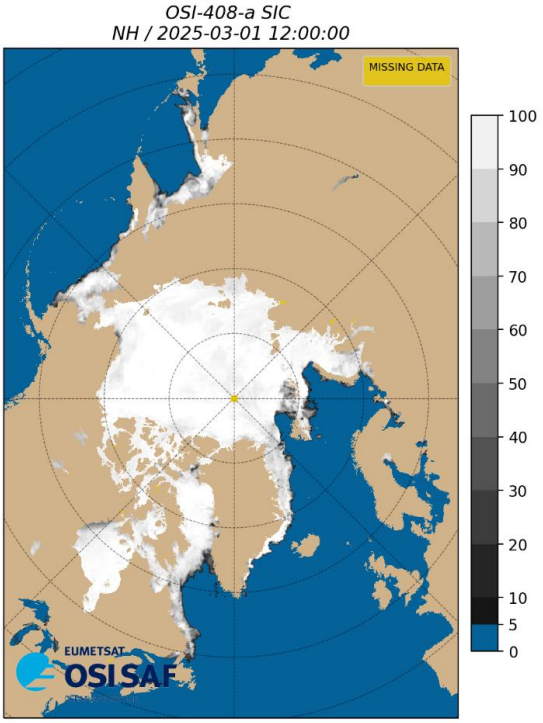


Figure 13: Example of the OSI SAF sea ice concentration product, OSI-401, for March 1st 2025.

Table 6: General information on dataset intersection in terms of time series and number of analysis (products, ice charts) with a timely and geographical overlap and how many grid cells/data points that correspond to (M = million).

Comparison datasets	*DMI regional ice chart and MET Svalbard ice chart*	*DMI overview ice chart and MET Svalbard ice chart*	*DMI overview ice chart and OSI SAF IC dataset*	*MET Svalbard ice chart and OSI SAF IC dataset*
Data time series	04-01-2010 -05-11-2013	02-03-2011 -05-11-2013	02-03-2011 -31-10-2013	04-01-2010 -31-10-2013
Analysis with overlap	118	98	231	938
Number of grid cells	9.5 M	66.4 M	2.5 M	16.4 M

Figure 14 and 15 show scatter density plots of DMI regional ice charts SIC versus MET ice charts SIC, and MET ice charts SIC versus DMI regional ice charts SIC, respectively. Black dots are mean values. Vertical lines are standard deviations. White numbers are occurrence in percent, rounded. Solid black line is a reference line (x=y). The corresponding Table 8 and 9 provide statistics based on the data shown in the corresponding figures.

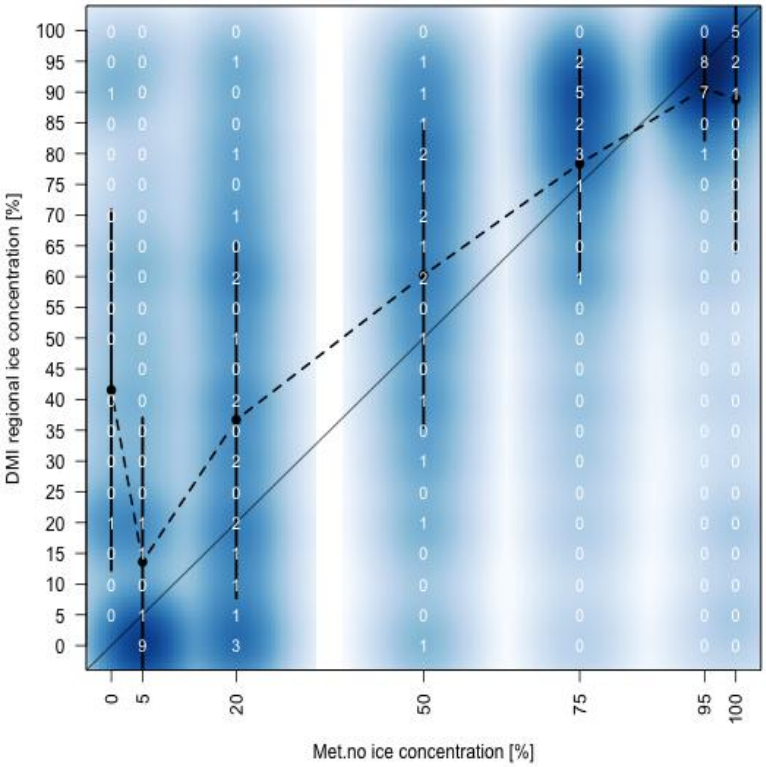


Figure 14: Scatter density plot MET Svalbard ice charts against DMI regional ice charts.

Table 7: Statistics based on the data also shown in figure 14.

MET	DMI		
IC	Mean	Std.dev.	Median
0	42	29	30
5	14	23	0
20	37	29	30
50	60	25	65
75	78	18	85
95	91	8	90
100	89	25	100

Figure 16 and 17 show Scatter density plots of [top] DMI overview ice charts IC and corresponding MET ice charts IC and [bottom] MET ice charts IC and corresponding DMI overview ice charts IC, 2011-2013. Black dots are mean values. Vertical lines are standard deviations. White numbers are occurrence in percent, rounded. Solid black line is a reference line (x=y). The corresponding Table 10 and 11 provide statistics based on the data also shown in the corresponding figures.
An additional analysis has been made, to investigate the difference in ice concentrations in an ice chart from each ice service, compared to the temporal evolution in ice concentrations of two subsequent ice charts from the same ice service. Here, the difference between ice concentrations in a MET ice chart and a DMI overview ice chart from the same date is compared to the difference in the corresponding ice concentrations values in two subsequent Greenland overview ice charts. See Figure 18.

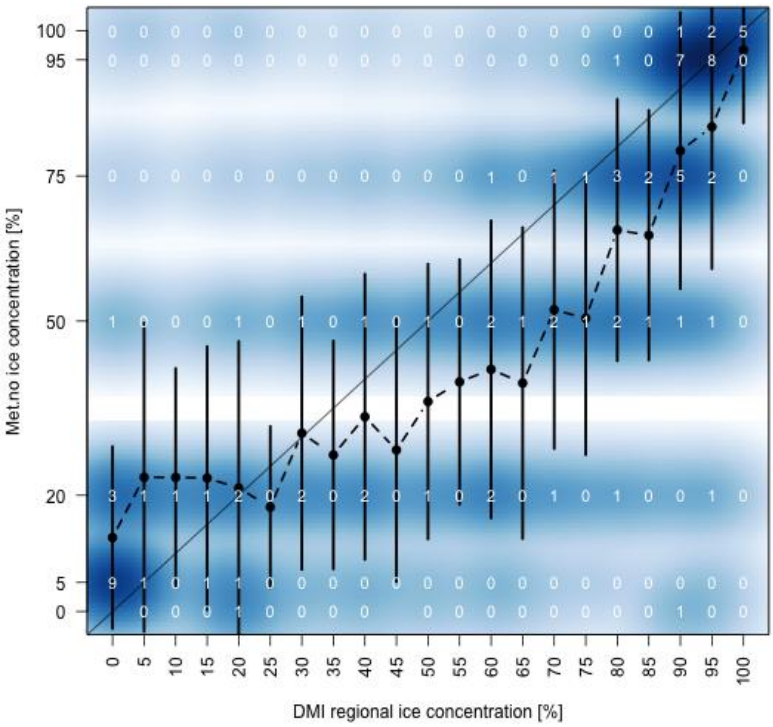


Figure 15: Scatter density plot DMI regional ice charts against MET Svalbard ice charts.

Table 8: Statistics based on the data also shown in Figure 15.

DMI	MET		
IC	Mean	Std.dev.	Median
0	11	14	5
5	25	28	20
10	24	21	20
15	22	22	20
20	19	25	5
25	17	14	20
30	30	22	20
35	28	20	20
40	31	22	20
45	24	21	20
50	38	23	50
55	38	20	50
60	42	25	50
65	38	25	50
70	48	22	50
75	51	24	50
80	66	22	75
85	62	21	75
90	76	27	75
95	78	28	95
100	97	12	100

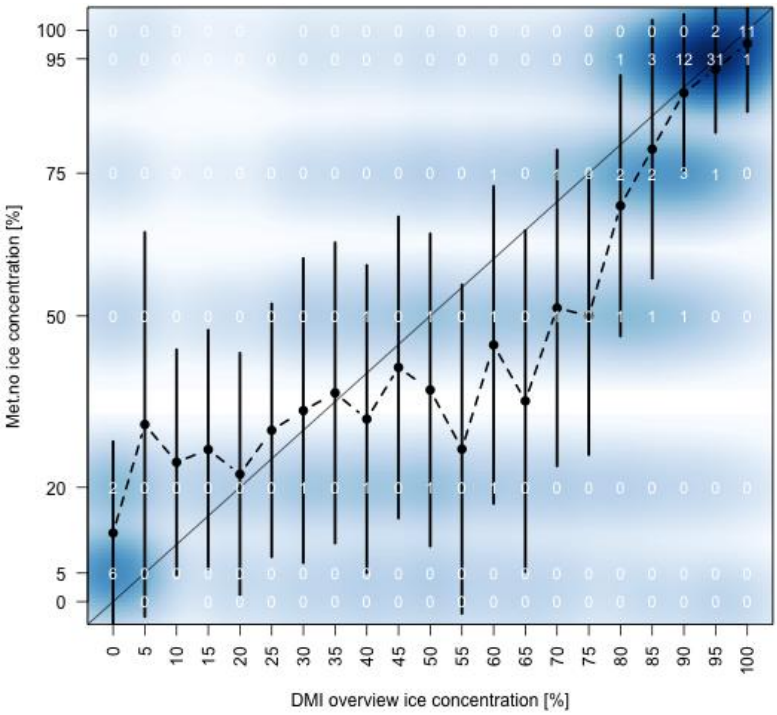


Figure 16: Scatter density plot DMI overview ice charts against MET Svalbard ice charts.

Table 9: Statistics based on the data also shown in Figure 16.

DMI	MET		
IC	Mean	Std.dev.	Median
0	12	16	5
5	31	34	20
10	24	20	20
15	27	21	20
20	22	21	20
25	30	22	20
30	33	27	20
35	37	26	20
40	32	27	20
45	41	26	50
50	37	27	20
55	27	29	20
60	45	28	50
65	35	30	20
70	51	28	50
75	50	24	50
80	69	23	75
85	79	23	95
90	89	14	95
95	93	11	95
100	98	12	100

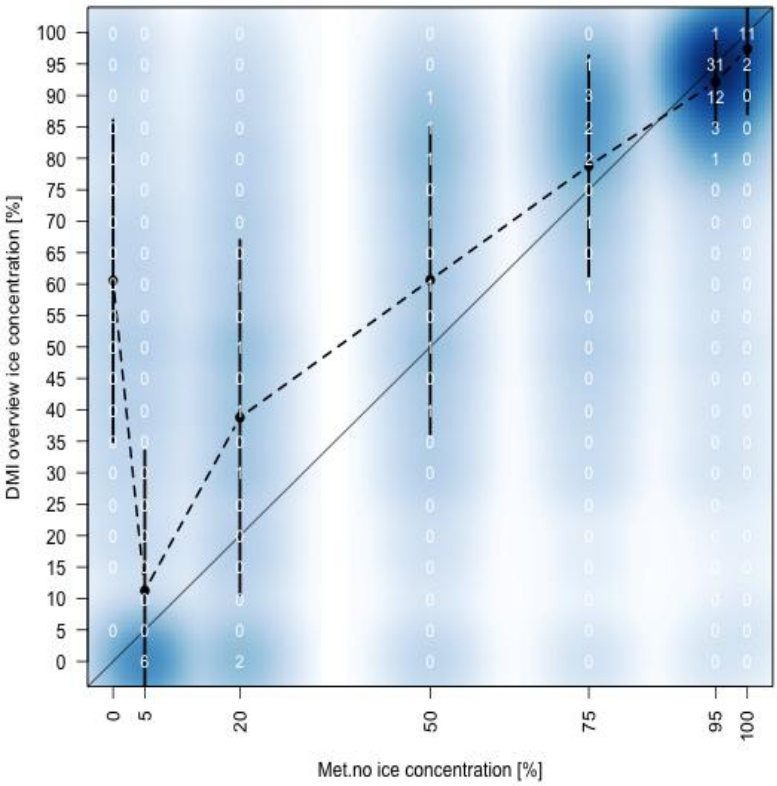


Figure 17: Scatter density plot MET Svalbard ice charts against DMI Overview ice charts.

Table 10: Statistics based on the data also shown in Figure 17.

MET	DMI		
IC	Mean	Std.dev.	Median
0	61	26	60
5	11	22	0
20	39	28	40
50	61	24	65
75	79	17	85
95	92	7	95
100	97	10	100

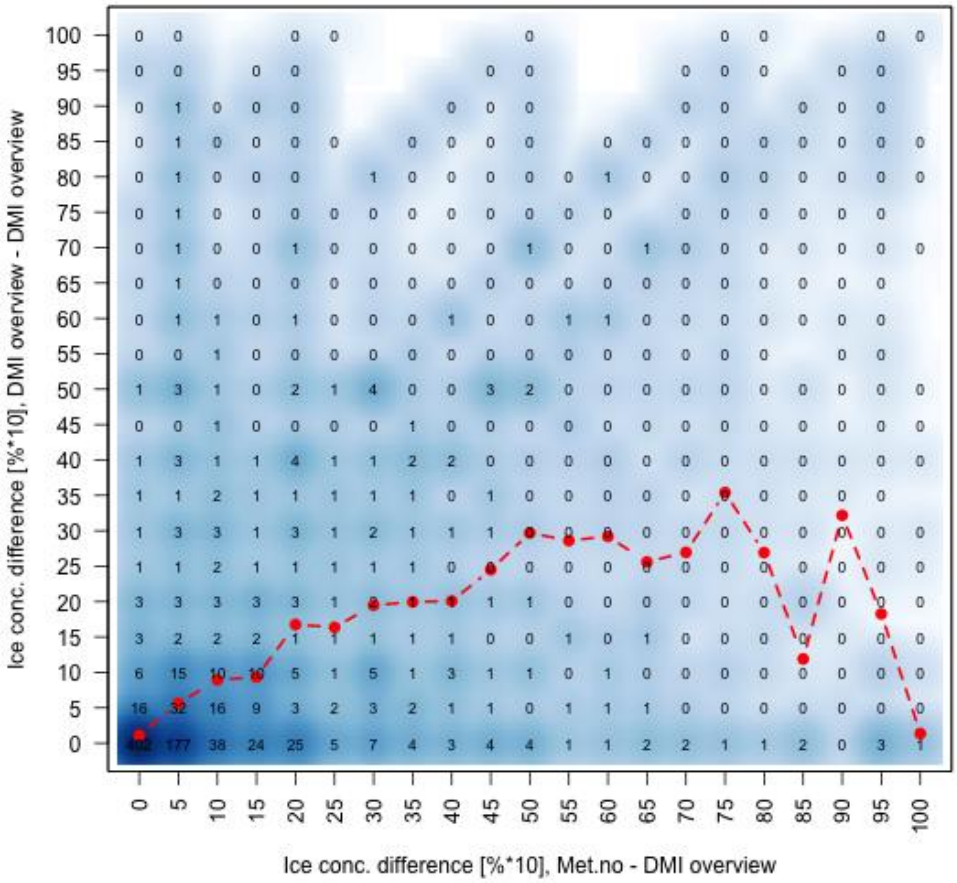


Figure 18: Scatter density plot of the IC difference between MET ice charts and corresponding DMI overview ice charts from the same day (t1) and the IC difference between two subsequent DMI overview ice charts (t1 to t2), for the period 2011-2013. Red dots are mean values. Black numbers are occurrence (of grid cell pairs) in percent times ten, rounded.

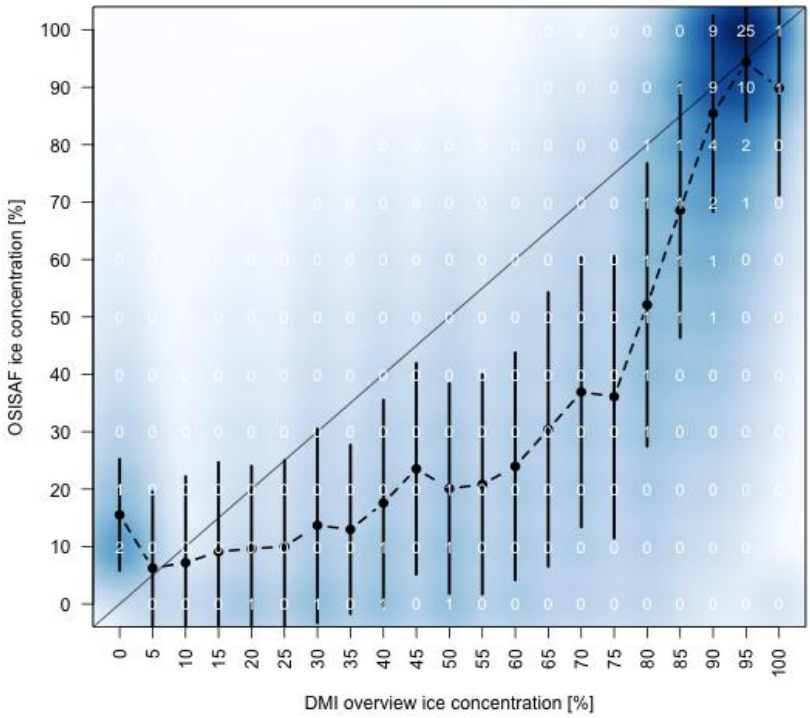


Figure 19: Scatter density plots of DMI overview ice charts IC and corresponding OSI SAF IC, 2011-2013. Black dots are mean values. Vertical lines are standard deviations. White numbers are occurrence in percent, rounded. Solid black line is a reference line (x=y).

Table 11: Statistics based on the data also shown in Figure 19.

DMI	OSI SAF		
IC	Mean	Std.dev.	Median
0	16	10	10
5	6	13	0
10	7	15	0
15	9	16	0
20	10	14	0
25	10	15	0
30	14	17	10
35	13	15	10
40	18	18	10
45	24	18	20
50	20	18	20
55	21	19	20
60	24	20	20
65	30	24	30
70	37	24	40
75	36	25	30
80	52	25	50
85	69	22	70
90	85	17	90
95	94	10	100
100	90	19	100

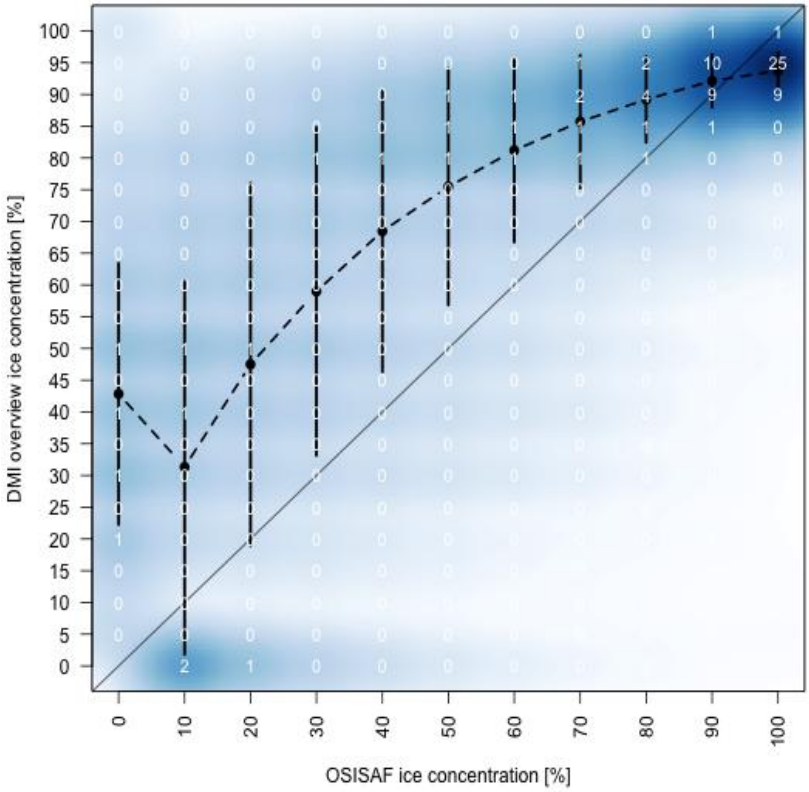


Figure 20: Scatter density plots of OSI SAF IC and corresponding DMI overview ice charts IC, 2011-2013. Black dots are mean values. Vertical lines are standard deviations. White numbers are occurrence in percent, rounded. Solid black line is a reference line (x=y).

Table 12: Statistics based on the data also shown in Figure 20.

OSI SAF	DMI		
IC	Mean	Std. dev.	Median
0	43	21	40
10	31	29	30
20	48	29	50
30	59	26	65
40	68	22	80
50	75	19	80
60	81	14	85
70	86	10	85
80	89	7	90
90	92	5	90
100	94	3	90

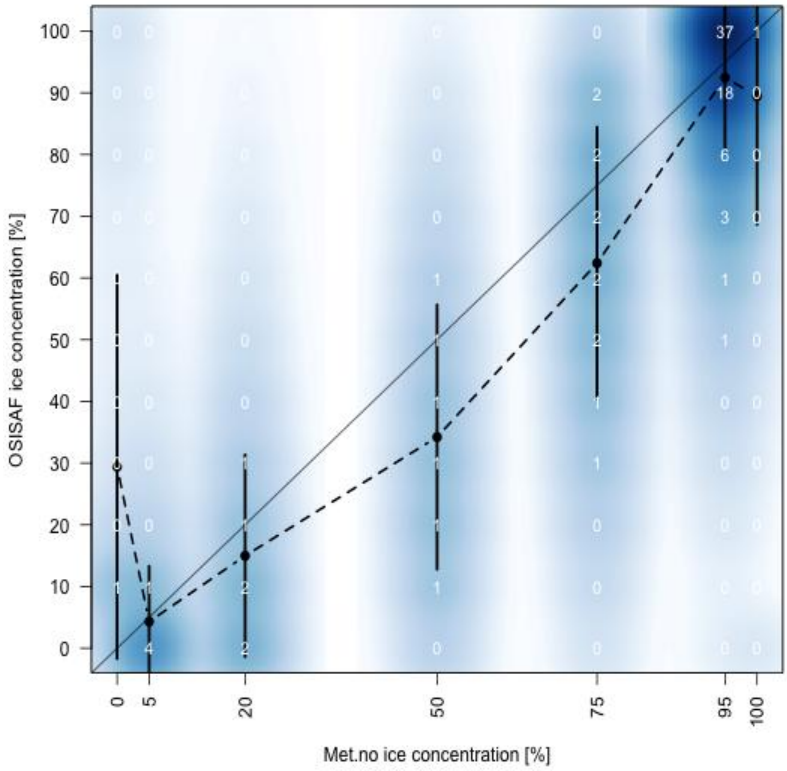


Figure 21: Scatter density plots of MET ice charts IC and corresponding OSI SAF IC, 2010-2013. Black dots are mean values. Vertical lines are standard deviations. White numbers are occurrence in percent, rounded. Solid black line is a reference line (x=y).

Table 13: Statistics based on the data also shown in Figure 21.

MET	OSI SAF		
IC	Mean	Std. dev	Median
0	29	31	10
5	4	9	0
20	15	16	10
50	34	21	30
75	62	22	70
95	92	11	100
100	89	21	100

Figure 19 and 20 show Scatter density plots of DMI overview ice charts IC versus OSI SAF product SIC and OSI SAF SIC versus DMI overview ice charts SIC, respectively. Figure 21 and 22 show scatter density plots of MET ice charts IC versus OSI SAF IC and OSI SAF IC versus MET ice charts IC, respectively. Black dots are mean values. Vertical lines are standard deviations. White numbers are occurrence in percent, rounded. Solid black line is a reference line (x=y). The corresponding Table 12 to 15 provide statistics based on corresponding Figures 19 to 22.

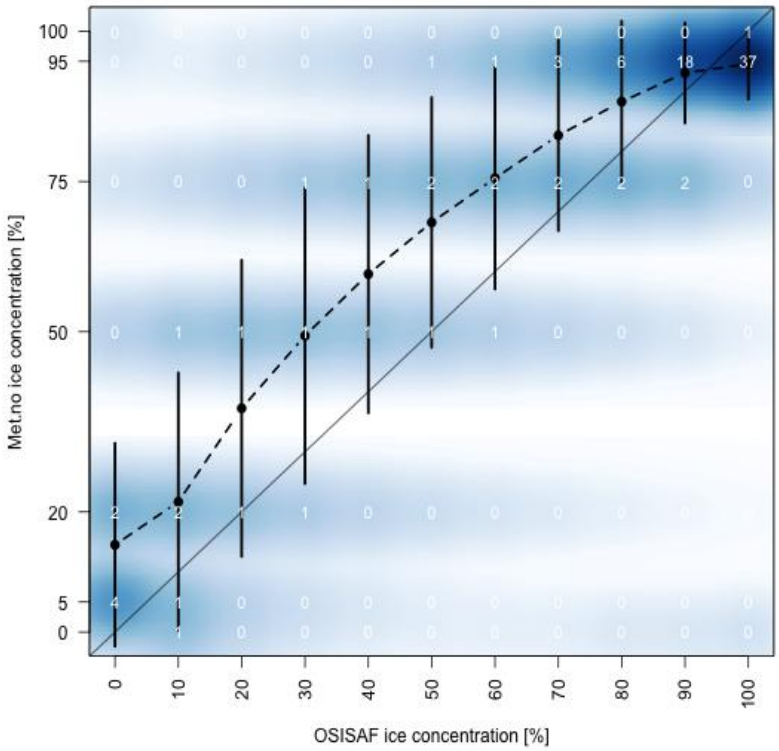


Figure 22: Scatter density plots of OSI SAF IC and corresponding MET ice charts IC, 2010-2013. Black dots are mean values. Vertical lines are standard deviations and white numbers occurrence in percent, rounded. Solid black line is a reference line (x=y).

Table 14: Statistics based on the data also shown in Figure 22.

OSI SAF	MET		
IC	Mean	Std. dev	Median
0	15	17	5
10	22	21	20
20	37	25	50
30	49	25	50
40	60	23	75
50	68	21	75
60	76	18	75
70	83	16	95
80	88	13	95
90	93	8	95
100	95	6	95

VIII.1.2 Discussion

The focus of this exercise was to investigate whether it is possible to develop common/standardized validation methods and derive uncertainty estimates and error statistics for the ice chart datasets.

The challenge of finding ice chart uncertainty estimates is to find a reasonable way to validate ice chart data sets and to identify all the different parameters affecting the data set quality. When trying to compute consistent and comparable uncertainty estimates for ice chart data sets, there are numerous factors controlling the quality that have to be taken into account.

- The history of changes in *data sources*, such as the increasing number of data sources for analysis, increasing resolution, better georeferencing and increasing volume of satellite input data.
- *Ice charting practices*, such as increasing knowledge of data interpretation and including the evolving analysis systems.

All these factors undoubtedly have had a positive influence on the quality of the ice chart data sets. Some of the factors that are harder to estimate are:

- Ice chart quality cannot always be measured on the *input imagery data quality*; since high resolution data sometimes can be more noise, due to weather phenomena, than lower resolution input imagery.
- The fact that ice charts are manually produced, introduces the element of the *Ice Analyst subjectivity*; ice chart data sets have a built-in inconsistency due to changing ice analysts, that cannot currently be measured.
- Most ice charts datasets are traditionally produced with a focus on *navigational awareness*.
- *Spatial scale* is a dominant factor when working with a parameter such as ice concentration, and though the ice chart data sets in MyOcean are assigned a resolution of 1 km, the factors described above states that the true resolution of the ice charts, the number and size of ice polygons, as well as the detail of the ice boundaries and the edge, are varying and inconsistent.

Some of the factors controlling ice chart quality, that introduces known differences between the ice chart data sets considered in this study, are:

- Difference in the *purpose of the dataset* and the *ice polygon information*; if the ice polygons are chosen with the focus of dissemination information of only ice concentration or both ice concentration *and* thickness. Also, the level of detail in which the ice polygon parameters are described are different, such as the span in IC intervals, or the use of the term *estimated* when addressing an ice parameter.
- Difference in the *temporal* perspective of the data set. Some ice charts are based on one single SAR image and are analyzed with the focus of distributing the ice chart NRT, where other ice charts are based on several SAR images and distributed with a fixed temporal frequency, with a focus on a more thorough and prolonged analysis of the data.
- Differences in the *spatial range* of the data sets (coverage area); some data sets have fixed ranges, others have varying range ('going after the ice') and to some extent also varying zoom level based on different geographical focus.

Some level of uncertainty is unfortunately introduced when the data sets are prepared for the MyOcean Catalogue:

- The IC intervals are, to some extent, averages of the original ice concentrations, as done when converting (gridding) the original ice chart ice polygon data to NetCDF files. Also, some additional information is lost in the conversion, such as the detail of the ice edge location as well as information of *estimated* ice parameters.

This exercise is complicated by the fact that a majority of the factors mentioned above change in time and that almost none of these factors are currently, and has never been, logged in a way, that their influence on the data set can be investigated and measured. Based on the documentation

available from the different Ice Services it is evident that these factors should be weighted differently for the different ice chart datasets. The Ice Services (and the ice charting software) are currently not at a point where it is possible to automatically record more of this information useful for error estimation during the ice charting process.

Considering the factors mentioned above, that are found to affect ice chart quality, the conclusion is that the differences between the ice chart datasets are too large to make it possible to develop common/standardized validation methods for all data sets. From the results of the comparison analyses made in this study we are able to say something general about the variance and standard deviation between the ice chart datasets, that can serve as valuable information on the realism of the data sets: for ice concentration the ice chart datasets are closer to each other at low and high concentrations, and the standard deviation is higher for the intermediate values.

VIII.1.3 Conclusion

Results of all the comparison analysis show that there is in general a good consistency between the datasets for the high IC above 80%. This is due to the common spatial domain of the two datasets covering the Greenland Sea area and the western Arctic Ocean, which are areas of high ice concentrations. There is also, to a lesser extent, a good consistency between the datasets when IC below 20%. IC differences and standard deviations are highest for the intermediate IC values.

For intermediate IC intervals both the DMI regional *and* overview ice concentrations are generally higher than MET ice concentrations. Another interesting feature is that MET ice charts tend to have low ice concentrations of 5% where the DMI ice charts have open water (0%). For the comparison analyses involving the MET ice charts, the level of the standard deviations on the intermediate ICs (5-85%) is comparable to the large span in the intermediate IC intervals used in the MET ice charts. A local comparison analysis (results not shown here) of DMI regional ice charts and MET ice charts in a Northeast Greenland subarea, where the MET ice charts havetheir main focus, showed no significant differences to the overall analysis.

It was also found that there is a general tendency towards larger ice concentration differences between the DMI overview and MET ice chart datasets compared to the difference between two subsequent DMI overview ice charts (3-4 days between). This suggests that the consistency within the DMI dataset is more dominant than the agreement between the two ice services, on the interpretation of ice concentrations. This is another way of measuring the difference between the two ice service dataset.

The comparison of OSI SAF ice conc. data with MET ice charts and DMI overview ice charts, respectively, show that there is a tendency of ice charts having 95% ice where OSI SAF data have 100% ice. In between, for all intermediate intervals, there is a clear trend of MET ICs being 5-15% higher and DMI overview ice concentrations being 15-30% higher than the corresponding OSI SAF ice concentrations. These results could both be a matter of scale, since the OSI SAF data set has a lower spatial resolution, as well as it could be an expression of the general difference between the passive microwave data set and the practices of ice charting.

Seasonal data analysis shows that in summertime data there are more corresponding ICs in the intermediate intervals than in wintertime. This is due to the general thinning of the ice cover in summer. For the wintertime data, when the ice cover is consolidating, the data sets very much 'agree' on the highest ICs (>95%). The analysis of yearly data shows no significant interannual tendencies. After the analysis was made, it was found that there is a small bug in the MET Svalbard ice chart data set, that is generated when converting the original ice chart polygon data to the NetCDF format for the MyOcean Catalog. This means that the intermediate IC intervals in the MET ice chart data are off by 5% ice concentration, such that the true intermediate IC intervals should be 25%, 55% and 80% (instead of 20%, 50% and 75%). Thus, in the results of the comparisons with MET ice charts, this error can explain 5% of the difference in the intermediate ice concentrations.